

# **LAPORAN PRAKTIKUM EAS**



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**PROGRAM STUDI INFORMATIKA  
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JAWA TIMUR**

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## Uji Robot BNU

### Code :

```
/* **** */
* Program : ROBOT BNU 4.0
* Sesuai Buku - Robot IoT
* **** */

#include <ESP8266WiFi.h>
#include <PubSubClient.h>

String Topic;
String Payload;

const char* ssid = "HOTSPOT@UPNJATIM.AC.ID";
const char* password = "belanegara";

#define IN_1 D4 // Maju
#define IN_2 D3 // Mundur
#define IN_3 D8 // Belok Kanan
#define IN_4 D7 // Belok Kiri

// #define mqttServer "broker.emqx.io"
// #define mqttPort 1883

#define mqttServer "broker.emqx.io"
#define mqttPort 1883

WiFiServer server(80);
WiFiClient espClient;
PubSubClient client(espClient);

void receivedCallback(char* topic, byte* payload, unsigned int length) {
    Serial.print("Message received: ");
    Serial.println(topic);
    Serial.print("payload: ");

    for (int i = 0; i < length; i++) {
        Serial.print((char)payload[i]);
    }
    Serial.println();

    /* we got '1' -> Maju */
}
```

```

if ((char)payload[0] == '1') {
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_1, LOW);
}

/* we got '2' -> Kiri */
if ((char)payload[0] == '2') {
    digitalWrite(IN_4, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_4, LOW);
    digitalWrite(IN_1, LOW);
}

/* we got '3' -> Kanan */
if ((char)payload[0] == '3') {
    digitalWrite(IN_3, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_3, LOW);
    digitalWrite(IN_1, LOW);
}

/* we got '4' -> Mundur */
if ((char)payload[0] == '4') {
    digitalWrite(IN_2, HIGH);
    delay(100);
    digitalWrite(IN_2, LOW);
}
}

void setup() {
    Serial.begin(115200);
    delay(10);
    pinMode(IN_1, OUTPUT);
    pinMode(IN_2, OUTPUT);
    pinMode(IN_3, OUTPUT);
    pinMode(IN_4, OUTPUT);

    digitalWrite(IN_1, LOW);

```

```
digitalWrite(IN_2, LOW);
digitalWrite(IN_3, LOW);
digitalWrite(IN_4, LOW);

// Connect to WiFi network
Serial.println();
Serial.println();
Serial.print("Connecting to ");
Serial.println(ssid);
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected");

server.begin();
Serial.println("Server started");

Serial.print("Use this URL to connect: ");
Serial.print("http://");
Serial.print(WiFi.localIP());
Serial.println("/");

// Connect to Server IoT (CloudMQTT)

client.setServer(mqttServer, mqttPort);
client.setCallback(receivedCallback);

while (!client.connected()) {
    Serial.println("Connecting to CCloudMQTT...");

    if (client.connect("ESP32Client" )) {

        Serial.println("connected");

    } else {
        Serial.print("failed with state ");
        Serial.print(client.state());
        delay(2000);
    }
}
```

```
}
client.subscribe("Maju");
client.subscribe("Kiri");
client.subscribe("Kanan");
client.subscribe("Mundur");
}

void loop() {

  client.loop();

  WiFiClient client = server.available();
  if (!client) {
    return;
  }

  Serial.println("new client");
  while(!client.available()){
    delay(1);
  }

  String request = client.readStringUntil('\r');
  Serial.println(request);
  client.flush();

  if (request.indexOf("/IN_1on") > 0) {
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_1, LOW);
  }
  if (request.indexOf("/IN_1off") > 0) {
    digitalWrite(IN_1, LOW);
  }
  if (request.indexOf("/IN_2on") > 0) {
    digitalWrite(IN_2, HIGH);
    delay(100);
    digitalWrite(IN_2, LOW);
  }
  if (request.indexOf("/IN_2off") > 0) {
    digitalWrite(IN_2, LOW);
  }
  if (request.indexOf("/IN_3on") > 0) {
    digitalWrite(IN_3, HIGH);
```

```

    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_3, LOW);
    digitalWrite(IN_1, LOW);
}
if (request.indexOf("/IN_3off") > 0) {
    digitalWrite(IN_3, LOW);
}
if (request.indexOf("/IN_4on") > 0) {
    digitalWrite(IN_4, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_4, LOW);
    digitalWrite(IN_1, LOW);
}
if (request.indexOf("/IN_4off") > 0) {
    digitalWrite(IN_4, LOW);
}

// Return the response
client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html");
client.println("");
client.println("<!DOCTYPE HTML>");
client.println("<html>");
client.println("<head>");
client.println("<meta name='apple-mobile-web-app-capable' content='yes'
/>");
client.println("<meta name='apple-mobile-web-app-status-bar-style'
content='black-translucent' />");
client.println("</head>");
client.println("<body bgcolor = \"#f7e6ec\">");
client.println("<hr/><hr>");
client.println("<h4><center> Robot BNU 4.0 </center></h4>");
client.println("<hr/><hr>");
client.println("<br><br>");
client.println("<br><br>");
client.println("<center>");
client.println("ROBOT");
client.println("<a href='\"/IN_1on\"'><button>Maju </button></a>");

```

```

client.println("<a href=\""/IN_2on\""/><button>Mundur </button></a><br
/>");
client.println("</center>");
client.println("<br><br>");
client.println("<center>");
client.println("ROBOT");
client.println("<a href=\""/IN_3on\""/><button>Belok Kanan
</button></a>");
client.println("<a href=\""/IN_4on\""/><button>Belok Kiri </button></a><br
/>");
client.println("</center>");
client.println("<br><br>");
client.println("<center>");
client.println("<table border=\"5\">");
client.println("<tr>");
//=====
if (digitalRead(IN_1))
{
    client.print("<td>Maju = ON</td>");
}
else
{
    client.print("<td>Maju = OFF</td>");
}
client.println("<br />");
//=====
if (digitalRead(IN_2))
{
    client.print("<td>Mundur = ON</td>");
}
else
{
    client.print("<td>Mundur = OFF</td>");
}
client.println("</tr>");
//=====
if (digitalRead(IN_3))
{
    client.print("<td>Belok Kanan = ON</td>");
}
else
{
    client.print("<td>Belok Kanan = OFF</td>");
}

```

```

    }
    client.println("<br />");
    //=====
    if (digitalRead(IN_4))
    {
        client.print("<td>Belok Kiri = ON</td>");
    }
    else
    {
        client.print("<td>Belok Kiri = OFF</td>");
    }
    client.println("</tr>");
    //=====

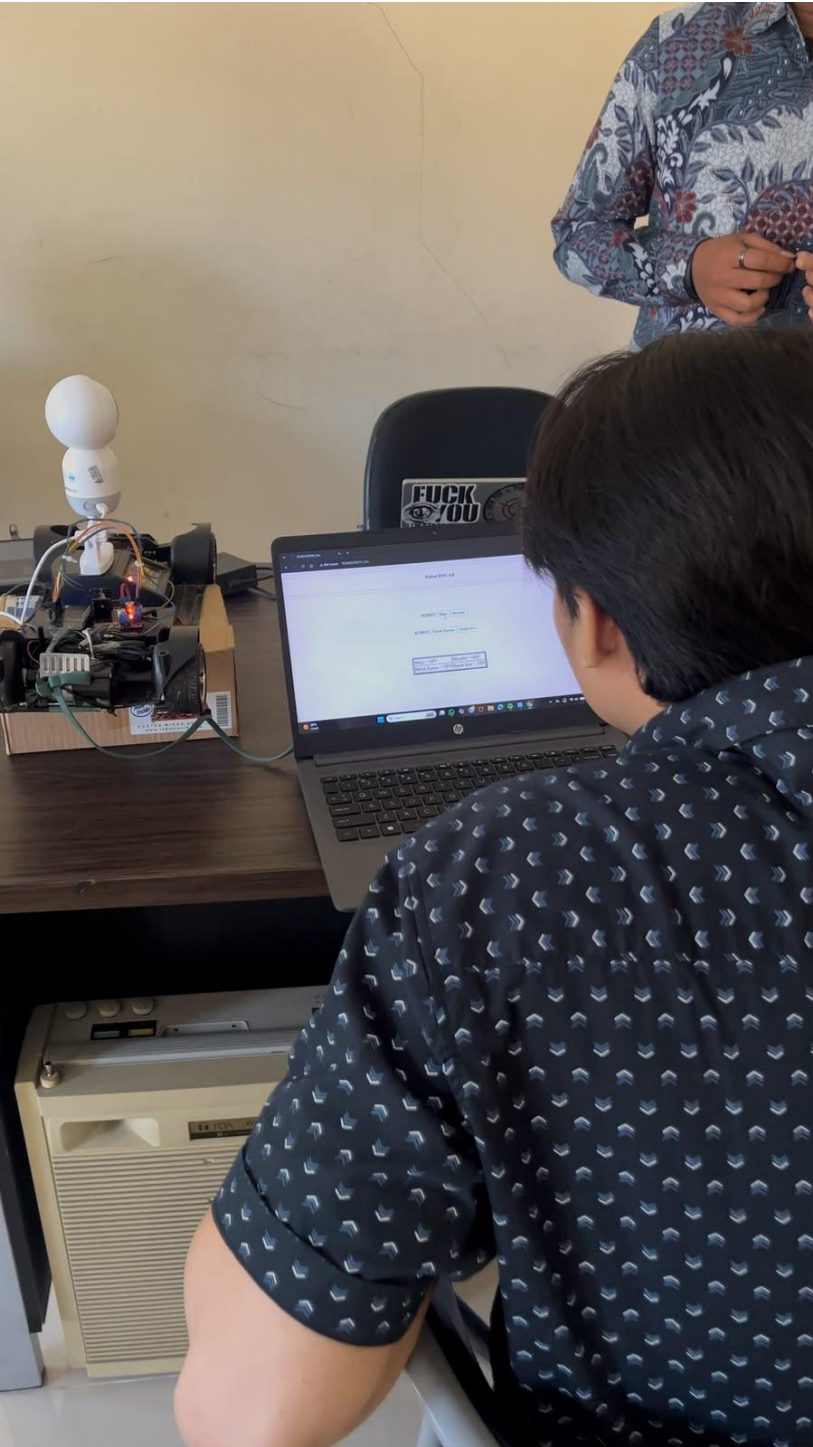
    client.println("<tr>");
    client.println("</table>");
    client.println("</center>");
    client.println("</html>");
    delay(1);
    Serial.println("Client disconnected");
    Serial.println("");
}

```

### Hasil Uji Robot BNU.

Program ini merupakan implementasi Robot IoT berbasis ESP8266 yang menggabungkan teknologi WiFi, Web Server, dan MQTT. Robot dapat dikendalikan baik melalui halaman web maupun aplikasi MQTT. Sistem ini memungkinkan kontrol robot dari jarak jauh selama perangkat terhubung ke jaringan internet atau WiFi yang sama. Cocok digunakan sebagai proyek pembelajaran Internet of Things (IoT), robotika, dan komunikasi MQTT.





## Uji Coba Robot BNU Menggunakan IOT MQTT

```
/*
*****
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  Serial.print("Message received: ");
  Serial.println(topic);
  Serial.print("payload: ");

  for (int i = 0; i < length; i++) {
    Serial.print((char)payload[i]);
  }
  Serial.println();

  /* we got '1' -> Maju */
  if ((char)payload[0] == '1') {
```

```

    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_1, LOW);
}

/* we got '2' -> Kanan */
if ((char)payload[0] == '2') {
    digitalWrite(IN_4, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_4, LOW);
    digitalWrite(IN_1, LOW);
}

/* we got '3' -> Kiri */
if ((char)payload[0] == '3') {
    digitalWrite(IN_3, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_3, LOW);
    digitalWrite(IN_1, LOW);
}

/* we got '4' -> Mundur */
if ((char)payload[0] == '4') {
    digitalWrite(IN_2, HIGH);
    delay(100);
    digitalWrite(IN_2, LOW);
}
}

void setup() {
    Serial.begin(115200);
    delay(10);
    pinMode(IN_1, OUTPUT);
    pinMode(IN_2, OUTPUT);
    pinMode(IN_3, OUTPUT);
    pinMode(IN_4, OUTPUT);

    digitalWrite(IN_1, LOW);
    digitalWrite(IN_2, LOW);

```

```
digitalWrite(IN_3, LOW);
digitalWrite(IN_4, LOW);

// Connect to WiFi network
Serial.println();
Serial.println();
Serial.print("Connecting to ");
Serial.println(ssid);
WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected");

server.begin();
Serial.println("Server started");

// Connect to Server IoT (CloudMQTT)

client.setServer(mqttServer, mqttPort);
client.setCallback(receivedCallback);

while (!client.connected()) {
    Serial.println("Connecting to CCloudMQTT...");

    if (client.connect("ESP32Clienttito" )) {

        Serial.println("connected");

    } else {
        Serial.print("failed with state ");
        Serial.print(client.state());
        delay(2000);
    }
}
client.subscribe("Maju");
client.subscribe("Kiri");
client.subscribe("Kanan");
client.subscribe("Mundur");
}
```

```
void loop() {

  client.loop();

  WiFiClient client = server.available();
  if (!client) {
    return;
  }

  Serial.println("new client");
  while(!client.available()){
    delay(1);
  }

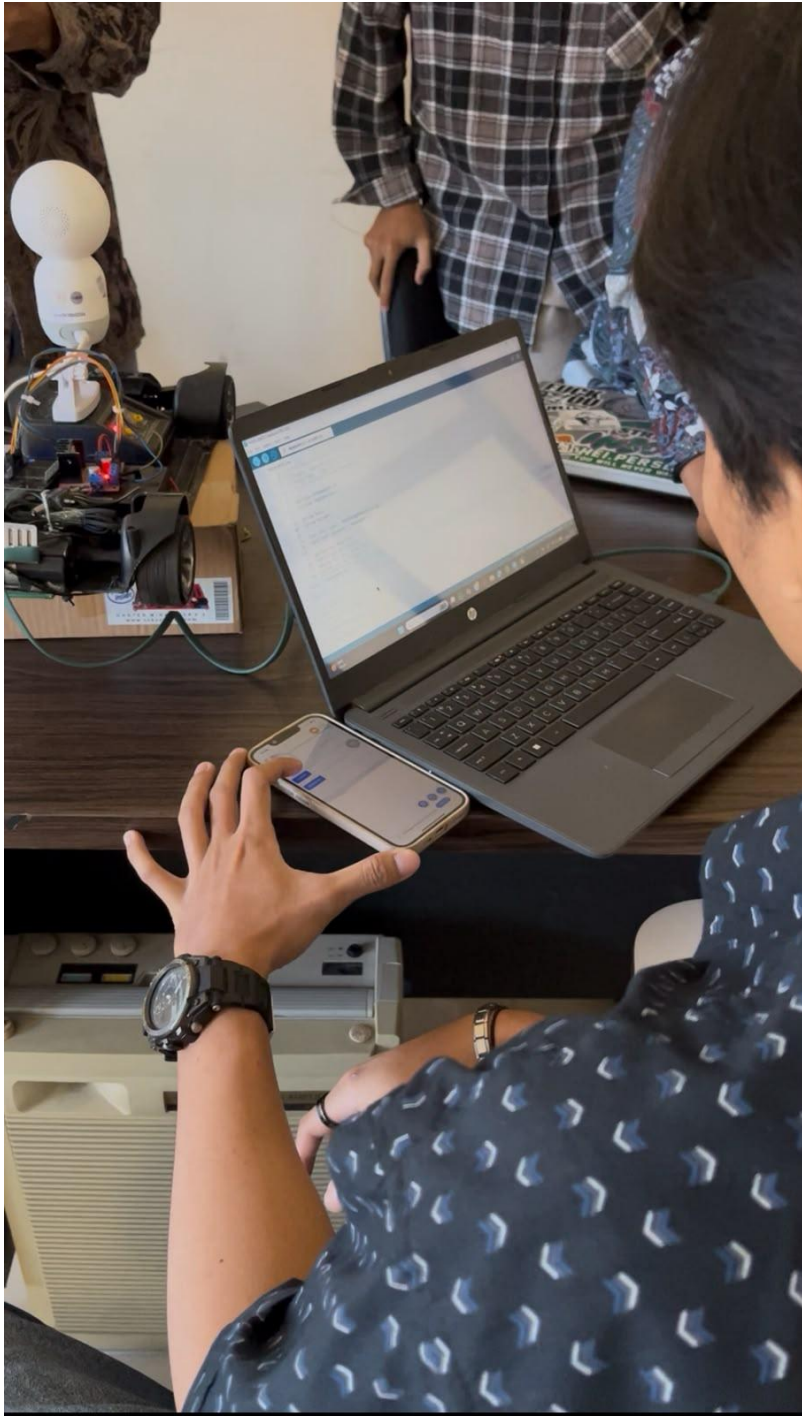
  String request = client.readStringUntil('\r');
  Serial.println(request);
  client.flush();

  if (request.indexOf("/IN_1on") > 0) {
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_1, LOW);
  }
  if (request.indexOf("/IN_1off") > 0) {
    digitalWrite(IN_1, LOW);
  }
  if (request.indexOf("/IN_2on") > 0) {
    digitalWrite(IN_2, HIGH);
    delay(100);
    digitalWrite(IN_2, LOW);
  }
  if (request.indexOf("/IN_2off") > 0) {
    digitalWrite(IN_2, LOW);
  }
  if (request.indexOf("/IN_3on") > 0) {
    digitalWrite(IN_3, HIGH);
    delay(300);
    digitalWrite(IN_1, HIGH);
    delay(100);
    digitalWrite(IN_3, LOW);
    digitalWrite(IN_1, LOW);
  }
}
```

```
if (request.indexOf("/IN_3off") > 0) {  
    digitalWrite(IN_3, LOW);  
}  
if (request.indexOf("/IN_4on") > 0) {  
    digitalWrite(IN_4, HIGH);  
    delay(300);  
    digitalWrite(IN_1, HIGH);  
    delay(100);  
    digitalWrite(IN_4, LOW);  
    digitalWrite(IN_1, LOW);  
}  
if (request.indexOf("/IN_4off") > 0) {  
    digitalWrite(IN_4, LOW);  
}  
}
```

### **Hasil Uji Coba Robot BNU Menggunakan IOT MQTT**

Berdasarkan pengujian, sistem Robot BNU 4.0 berhasil menerapkan konsep Internet of Things (IoT) dengan memanfaatkan ESP8266, WiFi, MQTT, dan Web Server. Robot mampu menerima perintah dari jaringan internet secara real-time dan menerjemahkannya menjadi aksi fisik berupa gerakan maju, mundur, belok kanan, dan belok kiri. Hasil pengujian menunjukkan bahwa komunikasi data antara pengguna, broker MQTT, dan perangkat IoT berjalan dengan baik sehingga sistem dapat digunakan sebagai media pembelajaran dan implementasi kontrol robot jarak jauh berbasis IoT.



## Hasil Connect Robolife Menggunakan SmartLife

Praktikum membaca IP Camera Robolife Smart IP Camera G1X2 menggunakan aplikasi Smartlife memulai dengan menghubungkan kamera ke jaringan WiFi, yang memungkinkan smartphone untuk mengakses perangkat. Aplikasi Smartlife menampilkan data seperti tanggal, waktu, kekuatan sinyal WiFi, dan kecepatan data saat ini setelah kamera terhubung. Antarmuka aplikasi menawarkan berbagai kontrol, seperti tombol untuk mengambil snapshot, mengaktifkan mikrofon untuk komunikasi dua arah, melihat rekaman playback, dan menu pengaturan untuk konfigurasi tambahan. Selain itu, ada kontrol arah berbentuk joystick virtual yang memungkinkan pengguna menggerakkan kamera ke kanan, kiri, atas, atau bawah sesuai kebutuhan. Tab navigasi seperti Messages, Cloud, dan Features tersedia di bagian bawah aplikasi. Tab-tab ini memungkinkan Anda mengakses fungsi tambahan seperti penyimpanan cloud atau notifikasi. Oleh karena itu, IP Camera Robolife Smartlife ini tidak hanya menampilkan video pengawasan, tetapi juga memberikan komunikasi, kendali penuh atas arah kamera, dan fitur tambahan untuk pengawasan. Ini membuat sistem pengawasan berbasis IoT lebih fleksibel dan interaktif.

